Math 22 Curriculum

Geometry | Middlesex School Mathematics | Resource: Math 22 Geometry Packet

COURSE AT A GLANCE

Review Content

REVIEW

No mapped OpenStax section

UNIT LEARNING OBJECTIVES

I can solve a linear equation arising from a geometric relationship.

I can solve a proportion and interpret the resulting scale factor.

I can simplify an exact square root and approximate it when needed.

I can calculate distance, midpoint, and slope on the coordinate plane.

I can substitute measurements into a formula with consistent units.

Geometry Basics

REQUIRED

1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

UNIT LEARNING OBJECTIVES

I can name and represent points, lines, planes, segments, and rays using correct notation.

I can identify collinear, coplanar, intersecting, and concurrent figures.

I can calculate segment lengths using betweenness and the segment-addition postulate.

I can determine a midpoint or use a segment bisector relationship.

I can classify and measure angles.

I can calculate angle measures using angle addition and angle bisectors.

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

I can distinguish a definition, postulate, theorem, converse, and counterexample.

I can determine angle relationships formed by parallel lines and a transversal.

I can use angle relationships to determine whether lines are parallel.

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

I can construct a logical geometric argument from stated facts and accepted results.

Triangle Congruence and Proof

REQUIRED

2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

UNIT LEARNING OBJECTIVES

I can classify triangles by sides and angles.

I can use triangle angle-sum and exterior-angle relationships.

I can write a congruence statement with corresponding vertices in the correct order.

I can determine whether triangles are congruent by SAS.

I can determine whether triangles are congruent by SSS.

I can determine whether right triangles are congruent by HL.

I can determine whether triangles are congruent by ASA or AAS.

I can apply isosceles and equilateral triangle theorems and their converses.

I can use corresponding parts of congruent triangles after proving congruence.

I can write a structured proof involving triangle congruence.

Quadrilaterals, Perimeter, and Area

REQUIRED

3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

UNIT LEARNING OBJECTIVES

I can calculate interior or exterior angle measures of a polygon.

I can calculate perimeter and area of common polygons.

I can apply properties of parallelograms to determine unknown measures.

I can determine whether a quadrilateral is a parallelogram.

I can apply properties of rectangles, rhombi, and squares.

Quadrilaterals, Perimeter, and Area (Continued)

REQUIRED

UNIT LEARNING OBJECTIVES - CONTINUED

I can classify a quadrilateral from its stated or marked properties.

I can apply trapezoid and trapezoid-midsegment relationships.

I can apply properties of kites.

I can calculate the perimeter or area of a composite polygon.

I can justify a quadrilateral conclusion using definitions and theorems.

Similarity and Proportional Reasoning

REQUIRED

4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

UNIT LEARNING OBJECTIVES

I can determine image lengths and coordinates under a dilation.

I can identify the scale factor of a dilation or similarity relationship.

I can write a similarity statement with corresponding vertices in the correct order.

I can use proportional corresponding sides to solve similar-polygon problems.

I can relate similarity scale factor to perimeter and area scale factors.

I can prove triangles similar by AA, SSS, or SAS.

I can apply triangle proportionality theorems and their converses.

I can solve and justify an indirect-measurement or other similarity application.

Right Triangles

REQUIRED

5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

UNIT LEARNING OBJECTIVES

I can apply the Pythagorean theorem to determine a missing side.

I can use the converse of the Pythagorean theorem to classify a triangle.

I can apply triangle-inequality relationships.

I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

I can apply right-triangle similarity relationships.

I can use geometric-mean relationships created by an altitude to the hypotenuse.

I can select sine, cosine, or tangent for a right-triangle problem.

I can determine a missing side or angle of a right triangle.

I can solve a contextual right-triangle problem.

I can justify a right-triangle solution using an appropriate theorem or ratio.

Circles

REQUIRED

6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

UNIT LEARNING OBJECTIVES

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

I can calculate circumference and arc length.

I can calculate the area of a circle, sector, or segment.

I can determine angle and arc measures involving central and inscribed angles.

I can apply relationships among chords, arcs, and distances from the center.

I can apply tangent-radius and tangent-segment relationships.

I can determine angle measures formed by chords, secants, or tangents.

I can determine segment lengths formed by intersecting chords.

I can determine segment lengths formed by secants or tangents.

I can solve a multi-step circle problem and justify each relationship used.

SKILL PROGRESSION I Introduce R Reinforce D Deepen A Apply

Review Content

REVIEW

Resource coverage: No mapped OpenStax section

M22-REV-001

I can solve a linear equation arising from a geometric relationship.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- A Check a proposed solution in the original equation.
  SK-ALG-CHECK-SOLUTION · first introduced in Math 12
- R Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12

M22-REV-002

I can solve a proportion and interpret the resulting scale factor.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- R Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- I Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR

M22-REV-003

I can simplify an exact square root and approximate it when needed.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- I Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX
- R Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21

M22-REV-004

I can calculate distance, midpoint, and slope on the coordinate plane.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- R Calculate horizontal or vertical distance from the difference of coordinates.
  SK-COORD-DISTANCE-AXIS · inherited horizontal/vertical coordinate distance
- I Calculate non-axis-aligned distance using the Pythagorean theorem or distance formula.
  SK-COORD-DISTANCE-PYTHAGOREAN
- R Calculate the midpoint of a segment from endpoint coordinates.
  SK-COORD-MIDPOINT · inherited from earlier mathematics
- R Calculate slope from two points.
  SK-LIN-SLOPE-POINTS · first introduced in Math 12

M22-REV-005

I can substitute measurements into a formula with consistent units.

Resource: No mapped OpenStax section

SUPPORTING SKILLS

- I Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS
- A Apply the order of operations to a numerical expression.
  SK-NUM-ORDER-OPERATIONS · inherited from middle school

Geometry Basics REQUIRED

Resource coverage: 1.1 Points, Lines, Planes; 1.2 Segments, Measurement, Addition, Midpoints, Bisectors; 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

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M22-BAS-001

I can name and represent points, lines, planes, segments, and rays using correct notation.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK
- I Recall definitions of points, lines, planes, segments, rays, angles, and intersections.
  SK-GEO-FOUNDATIONAL-DEFINITIONS
- I Read and write geometric notation correctly.
  SK-GEO-NOTATION

M22-BAS-002

I can identify collinear, coplanar, intersecting, and concurrent figures.

Resource: 1.1 Points, Lines, Planes

SUPPORTING SKILLS

- I Distinguish stated or marked facts from visual appearance.
  SK-GEO-DIAGRAM-INFER
- A Read and write geometric notation correctly.
  SK-GEO-NOTATION · first introduced in Math 22

M22-BAS-003

I can calculate segment lengths using betweenness and the segment-addition postulate.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Translate segment addition into an equation.
  SK-GEO-SEGMENT-ADDITION

M22-BAS-004

I can determine a midpoint or use a segment bisector relationship.

Resource: 1.2 Segments, Measurement, Addition, Midpoints, Bisectors

SUPPORTING SKILLS

- I Use a bisector to create equal measures.
  SK-GEO-BISECTOR
- I Use a midpoint to create equal segment lengths.
  SK-GEO-MIDPOINT
- A Translate segment addition into an equation.
  SK-GEO-SEGMENT-ADDITION · first introduced in Math 22

M22-BAS-005

I can classify and measure angles.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- I — Identify complementary, supplementary, vertical, and linear-pair relationships.
  SK-ANG-PAIR-IDENTIFY
- I — Measure or calculate an angle accurately.
  SK-GEO-ANGLE-MEASURE

M22-BAS-006

I can calculate angle measures using angle addition and angle bisectors.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I — Translate angle addition into an equation.
  SK-GEO-ANGLE-ADDITION
- A — Use a bisector to create equal measures.
  SK-GEO-BISECTOR · first introduced in Math 22

M22-BAS-007

I can identify and use complementary, supplementary, vertical, and linear-pair relationships.

Resource: 1.3 Angles, Measurement, Addition, Bisecting, Pairs; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- A — Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- D — Identify complementary, supplementary, vertical, and linear-pair relationships.
  SK-ANG-PAIR-IDENTIFY · first introduced in Math 22

M22-BAS-008

I can distinguish a definition, postulate, theorem, converse, and counterexample.

Resource: 1.1 Points, Lines, Planes; 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- I — Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE
- I — Disprove a universal statement with a counterexample.
  SK-PROOF-COUNTEREXAMPLE
- I — Apply a geometric definition as justification.
  SK-PROOF-DEFINITION
- I — Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT

M22-BAS-009

I can determine angle relationships formed by parallel lines and a transversal.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- **I** Use parallel-line angle relationships.
  SK-ANG-PARALLEL-APPLY
- **I** Identify corresponding, alternate, and same-side angle pairs.
  SK-ANG-TRANSVERSAL-IDENTIFY

M22-BAS-010

I can use angle relationships to determine whether lines are parallel.

Resource: 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- **I** Use angle relationships to prove lines parallel.
  SK-ANG-PARALLEL-CONVERSE
- **A** Identify corresponding, alternate, and same-side angle pairs.
  SK-ANG-TRANSVERSAL-IDENTIFY · first introduced in Math 22

M22-BAS-011

I can solve problems involving parallel, perpendicular, vertical, or horizontal lines.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular

SUPPORTING SKILLS

- **A** Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- **A** Use parallel-line angle relationships.
  SK-ANG-PARALLEL-APPLY · first introduced in Math 22
- **I** Use right-angle relationships created by perpendicular lines.
  SK-LIN-PERPENDICULAR-ANGLE

M22-BAS-012

I can construct a logical geometric argument from stated facts and accepted results.

Resource: 1.4 Reasoning; Angle/Line Pairs; Parallel and Perpendicular; 1.5 Parallel-Line Converses and Proofs

SUPPORTING SKILLS

- **I** Translate givens into diagram markings and statements.
  SK-PROOF-GIVEN
- **I** Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE
- **A** Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Triangle Congruence and Proof

REQUIRED

Resource coverage: 2.1 Triangles, Sides, and Angles; 2.2 Congruent Polygons; SAS; 2.3 SSS and HL; 2.5 ASA and AAS; 2.4 Isosceles and Equilateral Triangles

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M22-CON-001

I can classify triangles by sides and angles.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A — Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I — Classify a triangle by side lengths and angle measures.
  SK-TRI-CLASSIFY
- I — Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
  SK-TRI-CONGRUENCE-VOCAB

M22-CON-002

I can use triangle angle-sum and exterior-angle relationships.

Resource: 2.1 Triangles, Sides, and Angles

SUPPORTING SKILLS

- A — Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I — Apply the triangle angle-sum theorem.
  SK-TRI-ANGLE-SUM
- I — Apply the triangle exterior-angle theorem.
  SK-TRI-EXTERIOR-ANGLE

M22-CON-003

I can write a congruence statement with corresponding vertices in the correct order.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- I — Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE
- I — Write congruence statements in corresponding order.
  SK-PROOF-CONGRUENCE-ORDER
- A — Recall and correctly use terminology and notation for triangle classification, congruence, and corresponding parts.
  SK-TRI-CONGRUENCE-VOCAB · first introduced in Math 22

M22-CON-004

I can determine whether triangles are congruent by SAS.

Resource: 2.2 Congruent Polygons; SAS

SUPPORTING SKILLS

- A — Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D — Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- I — Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT

M22-CON-005

I can determine whether triangles are congruent by SSS.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-006

I can determine whether right triangles are congruent by HL.

Resource: 2.3 SSS and HL

SUPPORTING SKILLS

- A Use right-angle relationships created by perpendicular lines.
  SK-LIN-PERPENDICULAR-ANGLE · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-007

I can determine whether triangles are congruent by ASA or AAS.

Resource: 2.5 ASA and AAS

SUPPORTING SKILLS

- A Apply the triangle angle-sum theorem.
  SK-TRI-ANGLE-SUM · first introduced in Math 22
- D Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

M22-CON-008

I can apply isosceles and equilateral triangle theorems and their converses.

Resource: 2.4 Isosceles and Equilateral Triangles

SUPPORTING SKILLS

- D Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- I Apply isosceles-triangle theorems and converses.
  SK-TRI-ISOSCELES

M22-CON-009

I can use corresponding parts of congruent triangles after proving congruence.

Resource: 2.2 Congruent Polygons; SAS; 2.5 ASA and AAS

SUPPORTING SKILLS

- A Write congruence statements in corresponding order.
  SK-PROOF-CONGRUENCE-ORDER · first introduced in Math 22
- I Use corresponding parts only after triangle congruence is established.
  SK-PROOF-CPCTC

M22-CON-010

I can write a structured proof involving triangle congruence.

Resource: 2.3 SSS and HL; 2.4 Isosceles and Equilateral Triangles; 2.5 ASA and AAS

SUPPORTING SKILLS

- A Use corresponding parts only after triangle congruence is established.
  SK-PROOF-CPCTC · first introduced in Math 22
- A Translate givens into diagram markings and statements.
  SK-PROOF-GIVEN · first introduced in Math 22
- D Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- A Select SAS, SSS, HL, ASA, or AAS from marked information.
  SK-TRI-CONGRUENCE-SELECT · first introduced in Math 22

Quadrilaterals, Perimeter, and Area

REQUIRED

Resource coverage: 3.1 Polygon Angles, Perimeter, and Area; 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares; 3.4 Trapezoids; 3.5 Kites and Composite Polygons

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M22-QUAD-001

I can calculate interior or exterior angle measures of a polygon.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Calculate polygon interior or exterior angle sums.
  SK-POLY-ANGLE-SUM

M22-QUAD-002

I can calculate perimeter and area of common polygons.

Resource: 3.1 Polygon Angles, Perimeter, and Area

SUPPORTING SKILLS

- I Add boundary lengths without omitting or duplicating sides.
  SK-GEO-PERIMETER
- D Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22
- A Apply the order of operations to a numerical expression.
  SK-NUM-ORDER-OPERATIONS · inherited from middle school

M22-QUAD-003

I can apply properties of parallelograms to determine unknown measures.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A Translate quantities and relationships into an equation.
  SK-ALG-TRANSLATE-EQUATION · first introduced in Math 12
- I Apply parallelogram properties and tests.
  SK-QUAD-PARALLELOGRAM

M22-QUAD-004

I can determine whether a quadrilateral is a parallelogram.

Resource: 3.2 Parallelograms

SUPPORTING SKILLS

- A Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- D Apply parallelogram properties and tests.
  SK-QUAD-PARALLELOGRAM · first introduced in Math 22

M22-QUAD-005

I can apply properties of rectangles, rhombi, and squares.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- I Use the inclusive hierarchy of quadrilateral classifications.
  SK-QUAD-HIERARCHY
- I Distinguish and apply rectangle, rhombus, and square properties.
  SK-QUAD-RECT-RHOM-SQUARE

M22-QUAD-006

I can classify a quadrilateral from its stated or marked properties.

Resource: 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- A Distinguish stated or marked facts from visual appearance.
  SK-GEO-DIAGRAM-INFER · first introduced in Math 22
- I Recall the definitions and defining properties of the major quadrilateral families.
  SK-QUAD-DEFINITIONS
- D Use the inclusive hierarchy of quadrilateral classifications.
  SK-QUAD-HIERARCHY · first introduced in Math 22

M22-QUAD-007

I can apply trapezoid and trapezoid-midsegment relationships.

Resource: 3.4 Trapezoids

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Apply trapezoid and midsegment relationships.
  SK-QUAD-TRAPEZOID

M22-QUAD-008

I can apply properties of kites.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I Apply kite properties.
  SK-QUAD-KITE

M22-QUAD-009

I can calculate the perimeter or area of a composite polygon.

Resource: 3.5 Kites and Composite Polygons

SUPPORTING SKILLS

- I Decompose a composite region into familiar shapes.
  SK-GEO-AREA-DECOMPOSE
- D Add boundary lengths without omitting or duplicating sides.
  SK-GEO-PERIMETER · first introduced in Math 22
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-QUAD-010

I can justify a quadrilateral conclusion using definitions and theorems.

Resource: 3.2 Parallelograms; 3.3 Rhombi, Rectangles, Squares

SUPPORTING SKILLS

- D Apply a geometric definition as justification.
  SK-PROOF-DEFINITION · first introduced in Math 22
- A Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22

Similarity and Proportional Reasoning REQUIRED

Resource coverage: 4.1 Dilations and Similarity; 4.2 Similar Polygons, Perimeter, and Area; 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS; 4.5 Proportionality Theorems

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M22-SIM-001

I can determine image lengths and coordinates under a dilation.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- D Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-002

I can identify the scale factor of a dilation or similarity relationship.

Resource: 4.1 Dilations and Similarity

SUPPORTING SKILLS

- A Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- D Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22
- I Recall and correctly use the terms similarity, dilation, scale factor, and corresponding parts.
  SK-SIM-VOCAB

M22-SIM-003

I can write a similarity statement with corresponding vertices in the correct order.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- D Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- I Write similarity statements in corresponding order.
  SK-PROOF-SIMILARITY-ORDER

M22-SIM-004

I can use proportional corresponding sides to solve similar-polygon problems.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- A Use cross multiplication when two ratios form a proportion.
  SK-ALG-CROSS-MULTIPLY · inherited from middle school
- A Match corresponding vertices, sides, and angles in order.
  SK-GEO-CORRESPONDENCE · first introduced in Math 22
- I Set up proportions using corresponding sides.
  SK-SIM-PROPORTION

M22-SIM-005

I can relate similarity scale factor to perimeter and area scale factors.

Resource: 4.2 Similar Polygons, Perimeter, and Area

SUPPORTING SKILLS

- I Relate length scale factor to perimeter and area factors.
  SK-SIM-PERIMETER-AREA
- A Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

M22-SIM-006

I can prove triangles similar by AA, SSS, or SAS.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A Write similarity statements in corresponding order.
  SK-PROOF-SIMILARITY-ORDER · first introduced in Math 22
- D Organize statements and reasons into a valid proof sequence.
  SK-PROOF-STRUCTURE · first introduced in Math 22
- I Recognize AA triangle similarity.
  SK-SIM-AA
- I Test SSS or SAS triangle similarity.
  SK-SIM-SSS-SAS

M22-SIM-007

I can apply triangle proportionality theorems and their converses.

Resource: 4.5 Proportionality Theorems

SUPPORTING SKILLS

- A Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- I Apply triangle proportionality relationships.
  SK-SIM-TRI-PROPORTIONALITY

M22-SIM-008

I can solve and justify an indirect-measurement or other similarity application.

Resource: 4.3 Similarity by AA; 4.4 Similarity by SSS and SAS

SUPPORTING SKILLS

- A Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- D Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22
- A Calculate and interpret a similarity or dilation scale factor.
  SK-SIM-SCALE-FACTOR · first introduced in Math 22

Right Triangles

REQUIRED

Resource coverage: 5.1 Pythagorean Theorem and Triangle Inequalities; 5.2 Special Right Triangles; 5.3 HL Similarity and Right-Triangle Similarity; 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

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M22-RGT-001

I can apply the Pythagorean theorem to determine a missing side.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- **A** Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- **I** Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN

M22-RGT-002

I can use the converse of the Pythagorean theorem to classify a triangle.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- **A** Recognize and apply the converse of a conditional statement.
  SK-PROOF-CONVERSE · first introduced in Math 22
- **D** Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22

M22-RGT-003

I can apply triangle-inequality relationships.

Resource: 5.1 Pythagorean Theorem and Triangle Inequalities

SUPPORTING SKILLS

- **A** Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- **A** Classify a triangle by side lengths and angle measures.
  SK-TRI-CLASSIFY · first introduced in Math 22

M22-RGT-004

I can determine exact side lengths in 45-45-90 and 30-60-90 triangles.

Resource: 5.2 Special Right Triangles

SUPPORTING SKILLS

- **A** Distinguish exact values from measured or rounded approximations.
  SK-GEO-EXACT-APPROX · first introduced in Math 22
- **R** Extract perfect-power factors from a radical.
  SK-RAD-FACTOR-PERFECT · first introduced in Math 21
- **I** Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS

M22-RGT-005

I can apply right-triangle similarity relationships.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A Recognize AA triangle similarity.
  SK-SIM-AA · first introduced in Math 22
- D Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22

M22-RGT-006

I can use geometric-mean relationships created by an altitude to the hypotenuse.

Resource: 5.3 HL Similarity and Right-Triangle Similarity

SUPPORTING SKILLS

- A Set up proportions using corresponding sides.
  SK-SIM-PROPORTION · first introduced in Math 22
- I Apply geometric-mean relationships in a right triangle split by an altitude.
  SK-TRI-GEOMETRIC-MEAN

M22-RGT-007

I can select sine, cosine, or tangent for a right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Add congruence, parallel, perpendicular, and angle markings to a diagram.
  SK-GEO-DIAGRAM-MARK · first introduced in Math 22
- I Recall and correctly use the terms leg, hypotenuse, opposite side, adjacent side, and trigonometric ratio.
  SK-TRI-RIGHT-VOCAB
- I Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO
- I Label sides relative to a selected angle.
  SK-TRIG-TRIANGLE-LABEL

M22-RGT-008

I can determine a missing side or angle of a right triangle.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- I Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE
- I Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE

M22-RGT-009

I can solve a contextual right-triangle problem.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- A Restrict a model to meaningful contextual inputs.
  SK-MODEL-VALID-DOMAIN · first introduced in Math 12
- D Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22
- A Use an inverse trigonometric function to determine a missing angle.
  SK-TRI-TRIG-SOLVE-ANGLE · first introduced in Math 22
- A Use a trigonometric ratio to determine a missing side.
  SK-TRI-TRIG-SOLVE-SIDE · first introduced in Math 22

M22-RGT-010

I can justify a right-triangle solution using an appropriate theorem or ratio.

Resource: 5.4 Sine and Cosine; 5.5 Tangent and Solving Right Triangles

SUPPORTING SKILLS

- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22
- A Apply the Pythagorean theorem or its converse.
  SK-TRI-PYTHAGOREAN · first introduced in Math 22
- A Recall special-right-triangle ratios.
  SK-TRI-SPECIAL-RATIOS · first introduced in Math 22
- A Match opposite, adjacent, and hypotenuse sides to a trigonometric ratio.
  SK-TRI-TRIG-RATIO · first introduced in Math 22

Circles

REQUIRED

Resource coverage: 6.1 Circle Vocabulary, Circumference, Arc Length; 6.2 Circle, Sector, and Segment Area; 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

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M22-CIR-001

I can identify and name radii, diameters, chords, secants, tangents, arcs, sectors, and segments.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22
- A Read and write geometric notation correctly.
  SK-GEO-NOTATION · first introduced in Math 22

M22-CIR-002

I can calculate circumference and arc length.

Resource: 6.1 Circle Vocabulary, Circumference, Arc Length

SUPPORTING SKILLS

- I Relate a central angle to a fraction of a circle.
  SK-CIR-ARC-FRACTION
- I Calculate arc length from radius and central angle.
  SK-CIR-ARC-LENGTH
- I Calculate circumference from radius or diameter.
  SK-CIR-CIRCUMFERENCE
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-CIR-003

I can calculate the area of a circle, sector, or segment.

Resource: 6.2 Circle, Sector, and Segment Area

SUPPORTING SKILLS

- I Calculate the area of a circle.
  SK-CIR-AREA
- I Calculate sector area from radius and central angle.
  SK-CIR-SECTOR-AREA
- I Calculate segment area by subtracting a triangle from a sector.
  SK-CIR-SEGMENT-AREA
- A Select and report appropriate linear, square, or angular units.
  SK-GEO-UNITS · first introduced in Math 22

M22-CIR-004

I can determine angle and arc measures involving central and inscribed angles.

Resource: 6.3 Angle, Arc, and Chord Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Relate a central angle to a fraction of a circle.
  SK-CIR-ARC-FRACTION · first introduced in Math 22
- I Relate an inscribed angle to its intercepted arc.
  SK-CIR-INSCRIBED-ANGLE

M22-CIR-005

I can apply relationships among chords, arcs, and distances from the center.

Resource: 6.3 Angle, Arc, and Chord Relationships; 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I Apply equal-chord, equal-arc, and center-distance relationships.
  SK-CIR-CHORD-RELATION
- I Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB

M22-CIR-006

I can apply tangent-radius and tangent-segment relationships.

Resource: 6.4 Tangent, Secant, and Chord Relationships

SUPPORTING SKILLS

- I Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT
- I Use the perpendicular relationship between a radius and tangent.
  SK-CIR-TANGENT-RADIUS

M22-CIR-007

I can determine angle measures formed by chords, secants, or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- I Select the correct angle formula for chords, secants, or tangents.
  SK-CIR-ANGLE-RELATION
- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22

M22-CIR-008

I can determine segment lengths formed by intersecting chords.

Resource: 6.4 Tangent, Secant, and Chord Relationships; 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- I Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT

M22-CIR-009

I can determine segment lengths formed by secants or tangents.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Use inverse operations to maintain an equivalent equation.
  SK-ALG-INVERSE-OPERATIONS · first introduced in Math 12
- D Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22

M22-CIR-010

I can solve a multi-step circle problem and justify each relationship used.

Resource: 6.5 Tangent/Secant/Chord Angle, Arc, and Segment Relationships

SUPPORTING SKILLS

- A Select the correct angle formula for chords, secants, or tangents.
  SK-CIR-ANGLE-RELATION · first introduced in Math 22
- A Select and apply an intersecting-segment product relationship.
  SK-CIR-SEGMENT-PRODUCT · first introduced in Math 22
- A Identify circle parts and relationships from a diagram.
  SK-CIR-VOCAB · first introduced in Math 22
- D Select a theorem that connects known facts to a target statement.
  SK-PROOF-THEOREM-SELECT · first introduced in Math 22